

TE Mini Project Report

On

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Submitted in partial fulfillment of the requirement of
University of Mumbai for the Degree of

Bachelor of Engineering

in

Artificial Intelligence and Data Science.

Submitted By

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Department of Artificial Intelligence and Data Science.

Smt. Indira Gandhi College of Engineering, Ghansoli – 400701

An Autonomous Institute

Academic Year 2025 – 26



Department of Artificial Intelligence and Data Science.

SMT. INDIRA GANDHI COLLEGE OF ENGINEERING

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REPORT APPROVAL

This TE Mini Project report entitled “**Project Title**” by **Student Name1, Student Name2, Student Name3, and Student Name4** is approved for the degree of **Bachelor of Engineering** Department of Artificial Intelligence and Data Science, **Smt. Indira Gandhi College of Engineering, Ghansoli.** (An Autonomous Institute)

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Date:

Place:

Declaration

We declare that this written submission for TE Mini Project Report Declaration entitled “**Project Title**” represents our ideas in our own words and where others' ideas or words have been included. We have adequately cited and referenced the original sources. We also declared that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any ideas / data / fact / source in our submission. We understand that any violation of the above will cause disciplinary action by the institute and also evoke penal action from the sources which have thus not been properly cited or from whom paper permission has not been taken when needed.

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Table of Contents

Abstract.....	i
List of Figures.....	ii
List of Tables.....	iii
1. Introduction.....	1
1.1 Fundamentals.....	1
1.2 Objectives.....	1
1.3 Scope.....	2
1.4 Outline.....	2
2. Literature Survey.....	3
2.1 Introduction.....	3
2.2 Literature Review	4
2.3 Summary of Literature Survey.....	4
3. Proposed System(<i>Replace with Name of your model/application</i>).....	5
3.1 Overview.....	6
3.1.1 Existing System Architecture.....	8
3.1.2 Proposed System Architecture.....	9
3.2 Implementation Details	10

3.2.1	Algorithm / Tools/ Techniques	10
3.2.2	UseCase/Activity Diagram	10
3.2.3	Dataset and Input Details	11
3.2.4	Hardware and Software Specifications	11
4.	Performance and Results	13
4.1	Output and Graphs.....	14
4.2	Comparative Results	15
5.	Applications	16
5.1	Social	17
5.2	Technical	18
6.	Conclusion and Future Scope	19
References.....		
Acknowledgement.....		

Abstract

Here, try to write the abstract in 12 to 15 lines. An abstract is a self-contained, short, and powerful statement that describes a larger work. Components vary according to discipline. An abstract of a social science or scientific work may contain the scope, purpose, results, and contents of the work. An abstract is a brief summary of a work. It is often used to help the reader quickly ascertain the report's purpose and work done.

Give importance to your domain first. Mention techniques that are used in this domain. List what are the different possible techniques. Write what are limitations of other techniques? Write why a particular technique(s) is useful and used in this report. Describe your work in this report in 2 or 5 lines.

List of Figures

Figure 1.1	Title of the figure	1
Figure 2.1	ABC technique used by XYZ et al. [5]	4
Figure 3.1	Classification of domain techniques	6
Figure 3.2	Existing system architecture used for Content-based Systems	8
Figure 3.3	Proposed system architecture (if any)	9

List of Tables

Table 1.1	Title of the table	1
Table 2.1	Summary of literature survey	5
Table 3.1	Sample Dataset Used for Experiment	11
Table 3.2	Hardware details	11
Table 3.3	Software details	11

Chapter 1

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title of Chapter 1

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1.1 Fundamentals *(12pt Font, Times New Roman, Bold)*

This is the first Section Heading. The text in each section will be of the following format (Times New Roman 12, Justified, Line spacing 1.5). The same format is to be followed for all sections and subsections.

Page number must be in the footer of the page (starting from Introduction topic to references) and aligned to the centre of the page. The figures are centre aligned.

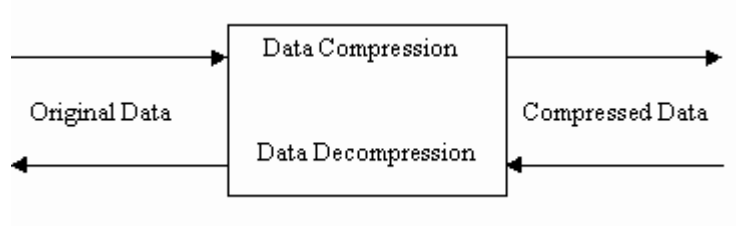


Fig 1.1: Title of the figure

The Figure 1.1 shows the method to add figure in the text. The description about the figure is to be written in one or two lines or more as per the requirements.

1.2 Objectives *(12pt Font, Times New Roman, Bold)*

The Table 1.1 shows the method to add table in the text. The description about the figure is to be written in one or two lines or more as per the requirements. The tables are centre aligned.

Table 1.1: Title of the table

Attribute 1	Attribute 2	Attribute 3
-------------	-------------	-------------

Value 1	Value 1	Value 1
Value 2	Value 2	Value 2

The objective is written in not more than 10 lines. The objective of this work is as follows:

1. To study the recommendation techniques and identify their limitations that may help to suggest a hybrid approach which may overcome the drawbacks existing methods.
2. To understand method of feature extraction for content based recommendation system and collaborative filtering that may help user for decisions making.
3. To identify evaluation metrics used for performance analysis of different recommendation systems.

1.3 Scope

Give here the scope of the project. Project scope is the part of project planning that involves determining and documenting a list of specific project goals, deliverables, tasks, costs and deadlines. The Project Scope pertains to the work necessary to deliver a product. Requirements and deliverables define the project scope. A scope statement should also include the list of users using the product, as well as the features in the resulting product.

1.4 Outline

The outline of the report is written in not more than 10 lines. The report is organized as follows: The introduction is given in Chapter 1. It describes the fundamental terms used in this project. It motivates to study and understand the different techniques used in this work. This chapter also presents the outline of the objective of the report. The Chapter 2 describes the review of the relevant various techniques in the literature systems. It describes the pros and cons of each technique. The Chapter 3 presents the Theory and proposed work. It describes the major approaches used in this work. The societal and technical applications are mentioned in Chapter 4. The summary of the report is presented in Chapter 5.

Chapter 2

Literature Survey

In this chapter the relevant techniques in literature is reviewed. It describes various techniques used in the work. Identify the current literature on related domain problem. Identify the techniques that have been developed and present the various advantages and limitation of these methods used extensively in literature.

Write advantage and disadvantage of this method. The summary of the literature presented at the end of this Chapter.

A literature review is an objective, critical summary of published research literature relevant to a topic under consideration for research. Its purpose is to create familiarity with current thinking and research on a particular topic, and may justify future research into a previously overlooked or understudied area [1].

A typical literature review consists of the following components:

1. Introduction
2. Main Body
3. Conclusion
4. References

2.1 Technique Category One

The techniques in this category are adapted to the individual needs, interests and preferences of user or society. They are tools for suggesting items to users in this domain. Various techniques in this category are listed here. These techniques have various advantages and are used extensively in literature.

2.1.1 Technique One

This technique is used by Author One [2] for XYZ purpose. It presents the state of the Art and discusses the main issues related to this domain problem. It discusses various techniques starting from simple techniques for representing structured data, to more complex techniques coming from the unstructured data. Write advantage and disadvantage of this method. This work is divided into three parts. Discuss each part here.

2.1.2 Technique Two

It was developed by Author Two et al. [3]. It presents the survey and experiments the two domain related techniques. These techniques show the complementary advantages and disadvantages. It compares the various techniques and shows which techniques are better based on the evaluation metrics. This fact has provided incentive for research in this domain for improved performance. It proposes various hybrid approaches that gives better results.

2.2 Technique Category Two

The techniques in this category are adapted to the individual needs, interests and preferences of user or society. They are tools for suggesting items to users in this domain. Various techniques in this category are listed here. These techniques have various advantages. Write advantage and disadvantage of this method.

2.2.1 Technique Three

Author Three et al. [4] applied similar techniques on different datasets. It introduces the flow of the domain system and the specific realization procedures of data input, data preprocessing, and other related steps. Results show that this technique in this domain has shown great performance.

2.2.2 Hybrid Approach One

Author Four has presented a hybrid approach [5] for ABC domain technique for XYZ purposed. The architecture is shown in Figure 2.1 It explores a new hybrid approach for solving the problem of finding the solutions based on user inputs. The proposed technique provides improvements in addressing two major challenges of this domain system: LMN and OPQ.

A					B
	X			Y	
		A	B		
		C	D		
	X			Y	
C					D

Fig. 2.1 ABC technique used by XYZ et al. [5]

The evaluation of the system shows superiority of the solution compared to stand-alone user-based ABC techniques. It shows that the hybrid approach improves the various problems.

2.3 Literature Summary

A literature review is an objective, critical summary of published research literature relevant to a topic under consideration for research. The summary is presented here.

Table 2.1 Summary of literature survey

SN	Paper	Advantages and Disadvantages
1.	Author One [2]	Advantages: Learning of profile is made easy. Quality improves over time. Considers implicit feedback. Disadvantages: Does not completely overcome the problem of abc and xyz.
2.	Author Two et al. [3]	Advantages: The web site based on this has shown great performance. Disadvantages: It does not recommend quality content to the users. Does not consider new user ABC problem. Not very efficient in terms of performance.
3.	Author Three et al. [4]	Advantages: The survey shows combine techniques for improved performance. It improves the user preferences for suggesting items to users.

4.	Author Four [5]	Advantages: It solves the problem of finding xyz. It improves lmn problem. Disadvantage: It does not consider the abc features which would give better results and solve the abc problem.
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Chapter 3

Proposed System

3.1 Overview

The system overview is presented in this Section. The classification of various techniques the domain is given in Figure 3.1

Fig. 3.1 Classification of domain techniques

This system helps user to find information by providing them with personalized suggestions. Based on above problems of researchers, recommendation techniques will have great influence in all aspects of our life.

Personalized Recommendation: The personalization is concern to adapting to the individual needs, interests and preferences of each user. It includes recommending, filtering and predicting techniques. From a business perspective, it is viewed as part of Customer Relationship Management (CRM) these techniques are content based recommendation system, collaborative filtering and knowledge based recommendation.

Non personalized recommendation: Non-personalized recommender systems are the simplest

type of recommender systems. As the name suggest, these types of recommender systems do not take into account the personal preferences of the users. The recommendations produced by these systems are identical for each customer. For example, if we go to amazon.com as an anonymous user it shows items that are currently viewed by other members. These systems recommend items to consumers based on what other consumers have said about the items or rated them on an average. Non-personalized recommender systems mainly use two types of algorithms: Aggregated opinion recommender and Basic product association recommender.

Aggregated opinion recommender: It displays the recommendations with a score which is an average of the ratings given to that specific thing by other customers. This score is basically a measure of how good the restaurant is. It scores generally range from 0 to 5. The average score calculated for most of such guides is as given in Equation 1.

$$\text{Score} = \text{round} (\text{MEAN} (\text{ratings}) * 10) \text{ or simply as } \text{Score} = \text{MEAN} (\text{ratings} * 10) \quad (3.1)$$

Averages are useful for an overall sense of what the population feels. However these averages lack context during recommendations. In above examples one bad rating or review has a lot of weight and can pull down an otherwise excellent rating. To tackle this issue, some websites tallied the percentage of people who rated a particular item “good” or “bad. This leads us to the concept of product association recommender.

Content Based Recommender Systems: Systems implementing a content based recommendation approach analyze a set of documents and/or descriptions of items previously rated by a user, and build a model or profile of user interests based on the features of the objects rated by that user. The profile is a structured representation of user interests, adopted to recommend new interesting items. The recommendation process basically consists in matching up the attributes of the user profile against the attributes of a content object. The result is a relevance judgment that represents the user’s level of interest in that object. If a profile accurately reflects user preferences, it is of tremendous advantage for the effectiveness of an information access process. For instance, it could be used to filter search results by deciding whether a user is interested in a specific Web page or not and, in the negative case, preventing it from being displayed.

3.1.1 Existing System Architecture

The Content Based Information Filtering (IF) systems need proper techniques for representing the items and producing the user profile, and some strategies for comparing the user profile with the item representation.

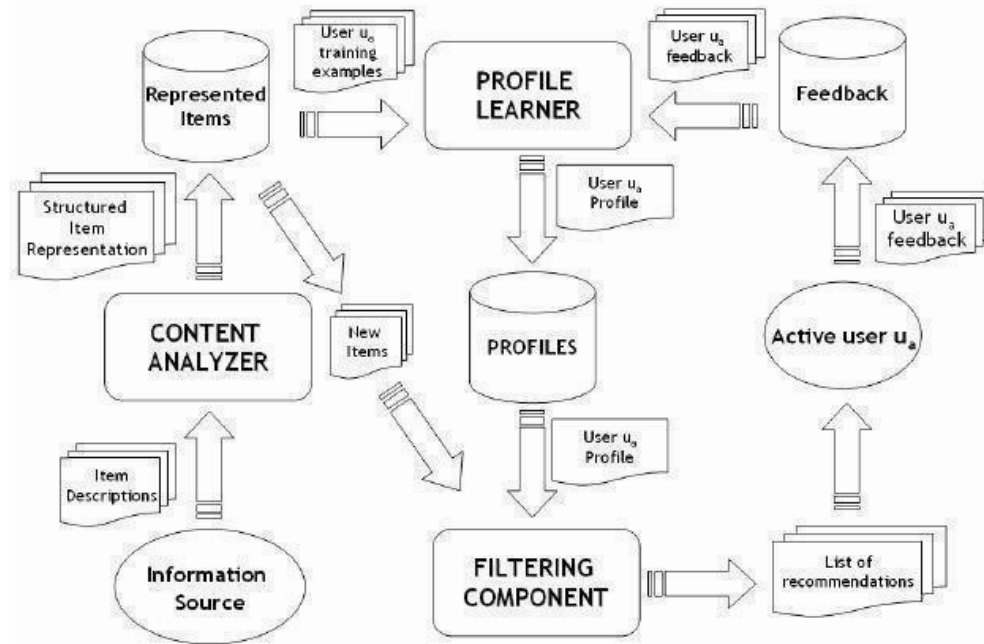


Fig. 3.1: High level architecture of a Content-based Recommender

Fig. 3.2 Existing system architecture used for Content- based Systems [2]

The high level architecture of a content- based recommender system is depicted in Figure 3.2. The recommendation process is performed in three steps, each of which is handled by a separate component:

- **Content Analyzer:** When information has no structure (e.g. text), some kind of pre-processing step is needed to extract structured relevant information. The main responsibility of the component is to represent the content of items (e.g. documents, Web pages, news, product descriptions, etc.) coming from information sources in a form suitable for the next processing steps.
- **Profile Learner:** This module collects data representative of the user preferences and tries to generalize this data, in order to construct the user profile. Usually, the generalization strategy is realized through machine learning techniques, which are able to infer a model

of user interests starting from items liked or disliked in the past. For instance, the profile learner of a Web page recommender can implement a relevance feedback method in which the learning technique combines vectors of positive and negative examples into a prototype vector representing the user profile.

- **Filtering Component:** This module exploits the user profile to suggest relevant items by matching the profile representation against that of items to be recommended. The result is a binary or continuous relevance judgment (computed using some similarity metrics), the latter case resulting in a ranked list of potentially interesting items.

3.1.2 Proposed System Architecture

The previous sections discussed the strengths and weaknesses of exiting system. In order to achieve better domain results, researchers combined both techniques to build Hybrid domain systems, which seek to inherit vantages and eliminate disadvantages.

In general, hybrid recommenders are systems that combine multiple recommendation techniques together to achieve a synergy between them. Although there exist a number of recommendation approaches that are practical to merge (i.e. Collaborative, Content-based, Demographic and Knowledge-based Recommender), our work will mainly focus on the combination of CF and CBF techniques. The proposed architecture is shown in Figure 3.3

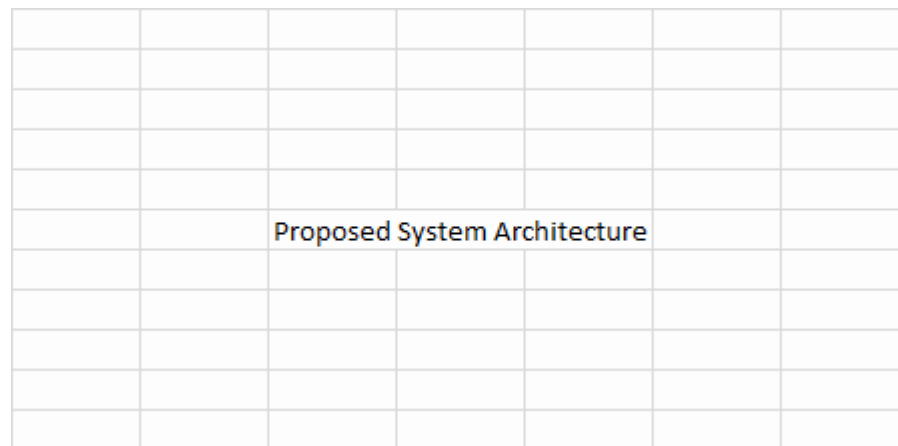


Fig. 3.3 Proposed system architecture (if any)

Depending on the domain and data characteristics, different types of combinations might produce dissimilar outputs. The following list describes several hybridization techniques that come into

consideration to merge CF and CBF recommenders

- **Weighted hybrid:** This hybrid combines scores from each component using linear formula. Therefore, components must be able to produce its recommendation score which can be linearly combinable. Also, the components have to be consistent relative accuracy across the product space and to perform uniformly.
- **Switching hybrid:** The issue of this hybrid is selecting one recommender among candidates. This selection is made according to the situation it is experiencing. The criterion for the selection like confidence value or external criteria should exist and the components might have different performance with different situations.
- **Mixed hybrid:** This is a hybrid which is based on the merging and presentation of multiple ranked lists into one. Each component of this hybrid should be able to produce recommendation lists with ranks and the core algorithm of mixed hybrid merges them into a single ranked list.
- **Feature combination hybrid:** There exist two very different recommendation components for this hybrid, contributing and actual recommender. The actual recommender works with data modified by the contributing one. The contributing one injects features of one source to the source of the other component.
- **Feature augmentation hybrid:** This is similar to the feature combination hybrids but different that the contributor generates new features. It is more flexible and adds smaller dimension than feature combination method.
- **Cascade hybrid:** This one is a tie breaker. The secondary recommender is just a tie breaker and does refinements.
- **Meta-level hybrid:** For this one, contributing and actual recommenders exist but the former one completely replaces the data for the latter one, not just part of it

Hybrid models discussed two techniques that have merged in four different ways. Write about it.

3.2 Implementation Details

The implementation detail is given in this section.

3.2.1 Technique ABC

Explain the techniques to be implemented with example. The procedure in detail is given in this Section. An algorithm shouldn't have computer code. Instead, the algorithm should be written in

such a way that, it can be used in similar programming languages. Algorithms are a way to communicate about your problem/solution with other people. Also it is a possible way to solve a given problem.

3.2.2 Use Case Diagram / Activity Diagram

Explain the diagram if any. Explain each block in at least one line with possible example. The procedure is given as follows. Give steps.

3.2.3 Output and Dataset

An experiment is conducted in order to identify the input/output behavior of the system. Identify inputs. Specify the sample inputs that would be used in the experiments. The sample dataset used in the experiment are identified and given in Table 3.1

Table 3.1 Sample Dataset Used for Experiment

Dataset	Users	Items	Interactions	Type
Filmtipset	81,282	64,747	19,812,490	Information
Last.fm 10, 681	359, 349	160, 154	3, 778, 552, 887	Personalized radio
Movielens	71, 567	10,681	10, 000, 054	Information

3.2.4 Hardware and Software Specifications

The experiment setup is carried out on a computer system which has the different hardware and software specifications as given in Table 3.2 and Table 3.3 respectively.

Table 3.2 Hardware details

Processor	2 GHz Intel
HDD	180 GB
RAM	2 GB

Table 3.3 Software details

Operating System	Windows XP Professional With Service pack 2
Programming Language	JDK 1.8
Database	Oracle 9

Chapter 4

Performance and Results

4.1 Graphs:

(explain/discuss the results with respect to figures)

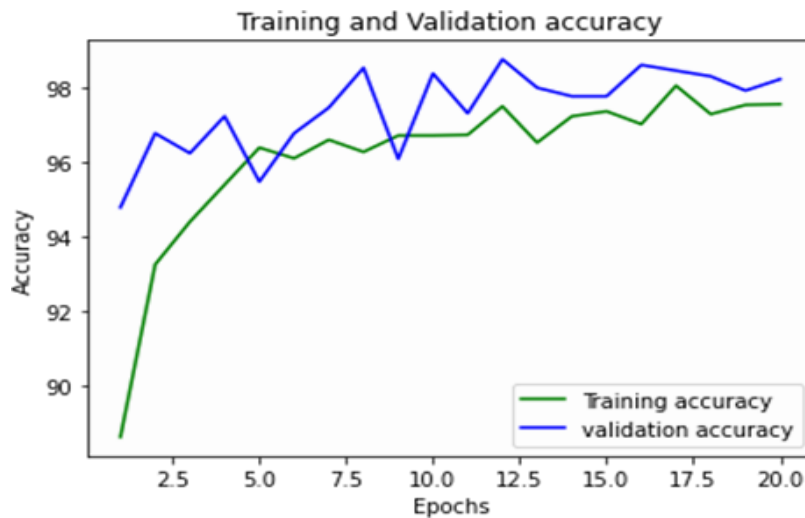


Fig 4.1 Graph showing increase in accuracy as number of epochs is increased

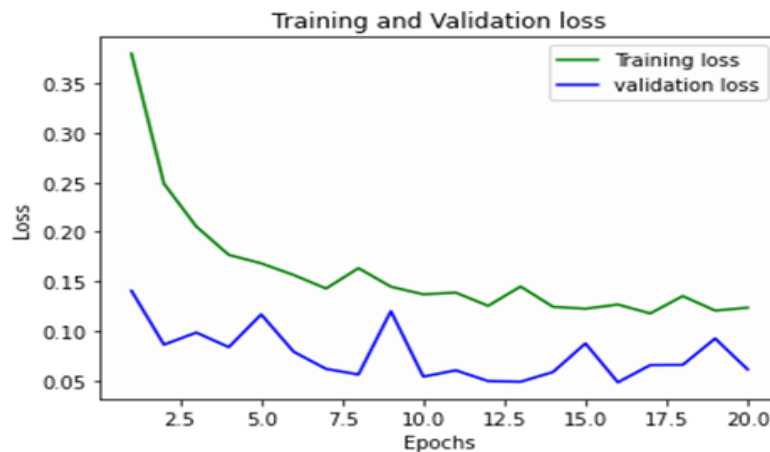


Fig 4.2 Graph showing decrease in loss as number of epochs is increased

Table 4.3 Confusion Matrix for our model

True positive = 1134	False positive = 8
False negative = 15	True negative =147

4.2 Comparative Results

Add screen shots of results.

Output Details and Screenshots

Chapter 5

Applications

There are various applications of this domain system. The application is listed here.

5.1 Social

Broadcast News Audio

Broadcasting news is considered as an important and interesting application field of content based audio classification systems. Since this system is used for segmenting audio streams into expressive regions, it has received special attention. For detection of particular event or file audio classification is used. The aim of using audio classification in news broadcasting is classifying recorded sound into pure speech and the other specified classes. So, the news can be smoothly broadcasted without any interruption.

Information Retrieval

The semantic retrieval of information is one of the applications in this field. The content based information retrieval has been developed using standards e.g. MPEG-7 [9]. For retrieving the data from the database, it is necessary to classify the data into similar type. So, it can be easily retrieved by the user. It is an easy task for user who might not have the knowledge about how the information is stored in the database.

5.2 Technical

Video Indexing

The audio classification system can be used in video indexing. The video is classified into different classes containing distinct features or properties with the help of classifier. So, the user can access the particular video by using the index or class of that video.

5.4 Content Analysis

Content based classification system can be used for analysis of audio/video content. Audio classification is also used in speaker identification for analysis of content [10]. In video/audio content, the number of speakers is undefined. So, in order to identify the particular speaker in the

given content audio classification is used. Also it is applied to analyze the content whether noise is present in content.

Dubbing of audio

The content based audio classification system can be used in dubbing of audio. While dubbing it is necessary to classify it, because an audio contains different types of sound. So in order to identify sound of a particular class, audio classification is used.

Multimedia Content Retrieval

The classification is widely used concept for multimedia retrieval. Multimedia data consists of images, texts, audios, and videos. It is necessary to arrange these classes in form of clusters of same type. So, the user can easily retrieve the multimedia data according to his/her choice. The multimedia data has been stored in the database, so the user can easily access that data.

Searching for Particular Sound

Searching for particular sound is the most commonly used application of content based audio classification. The audio data is classified into different classes or sound. So, the user can easily search a particular sound or class. It can also be used to identify different sounds like barking of dog, laughing etc.

Chapter 6

Conclusion and Future Scope

In this report, the study of different domain techniques is presented. The different techniques such as ABC, XYZ, content based, and collaborative filtering is explained with examples. The different hybrid approaches are also described. The comparative study of various techniques mentioned above is presented in this report. The hybrid approach is proposed with xyz modification. The performance measures like precision and recall are described in this report. The different standard datasets or variable inputs are defined that may be used in experiment for this domain system. The two datasets identified for experiments are CDEF and JKLM. The applications of this domain are identified and presented.

References

(The following text shall be 12pt Font, Times New Roman, Justified, 1.5 line spacing)
(All papers/theses/articles/letters/reports/urls used for writing project synopsis has to be mentioned in the Reference Section (here) and must to be cited in the report text wherever referred. For more details refer the url <https://www.ieee.org/documents/ieeecitationref.pdf>)

- [1] Pamela Fry (Thompson Rivers University). Literature Review Template [Online]. Available FTP: https://www.tru.ca/_shared/assets/Literature_Review_Template30564.pdf
- [2] J. K. Author, "Name of paper," Abbrev. Title of Periodical, vol. x, no. x, pp. xxx-xxx, Abbrev. Month, year.
- [3] R. E. Kalman, "New results in linear filtering and prediction theory," J. Basic Eng., ser. D, vol. 83, pp. 95-108, Mar. 1961.
- [4] A. B. Author. (year). Title (edition) [Type of medium]. Available FTP: Directory: File:
- [5] R. J. Vidmar. (1994). On the use of atmospheric plasmas as electromagnetic reflectors [Online]. Available FTP: atmnext.usc.edu Directory: pub/etext/1994 File: atmosplasma.txt.
- [6] A. B. Author, "Title of paper," in Unabbreviated Name of Conf., City of Conf., Abbrev. State (if given), year, pp. xxx-xxx.

Acknowledgement

(The following text shall be 12pt Font, Times New Roman, Justified, 1.5 line spacing)

We would like to express our special thanks to **Prof. Name of Guide**____, our mini project guide who guided us through the project and who helped us in applying the knowledge that we have acquired during the semester and learning new concepts.

We would like to express our special thanks to **Dr. Shankar M. Patil** Head, Department of Department of Artificial Intelligence and Data Science, who gave us the opportunity to do this mini project because of which we learned new concepts and their application.

We are also thankful to our mini project coordinator **Prof. Nirosha Uppu** along with other faculties for their encouragement and support.

Finally we would like to express our special thanks of gratitude to Principal **Dr. Sunil Chavan** who gave us the opportunity and facilities to conduct this mini project.

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Last paragraph may be used for others/family.
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Project member name 1
Project member name 2
Project member name 3
Project member name 4